

sncosmo_98bw_model

January 20, 2026

```
[1]: import redback
import numpy as np
import matplotlib.pyplot as plt
from astropy.cosmology import Planck18 as cosmo
import astropy.units as uu
from redback.transient_models.supernova_models import lambda_to_nu
import redback.interaction_processes as ip
import redback.sed as sed
import redback.photosphere as photosphere

import astropy.units as uu

import extinction
from extinction import ccm89, fitzpatrick99, apply, remove
#from scipy.interpolate import RegularGridInterpolator
import sncosmo
```

No module named 'lalsimulation'

lalsimulation is not installed. Some EOS based models will not work. Please use bilby eos or pass your own EOS generation class to the model

16:54 bilby INFO : Running bilby version: 2.3.0

16:54 redback INFO : Running redback version: 1.12.1

```
[2]: from redback.model_library import all_models_dict

func = all_models_dict['sncosmo_models']
```

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[3]: #1. Load in GRB event information:
#check 98bw first to see if it returns with 1
#use data from Jillian's file and constrain to B band only

AB_mag = 14.370000000000001 - 0.09 #B band correction
#have to use two frequencies or two time points - pass in the same time point
↳ twice -- does this get the code working --YES
target_obs_times = np.array([10.420000000074506, 10.420000000074506]) #days in
↳ observer frame
target_redshift = 0.0085
target_frequencies = 6.86300e+14 # B band (FROM REDBACK FILTERS)
```

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wavelength = np.array([4371.07000]) #angstroms
target_milky_way_ebv = 0.059 #from literature

#interpolate sncosmo model
baseline_flux = func(
    time=target_obs_times,
    redshift=target_redshift,
    model_kwargs=None,
    sncosmo_model='v19-1998bw-corr',
    output_format='flux_density', # Returns mJy
    frequency=target_frequencies,
    mw_extinction=True,           # Add MW dust for YOUR target
    ebv_mw=target_milky_way_ebv, # Specific to your GRB location
    # ADD THESE TWO LINES:
    model_kwarg_names=[], # This tells redback there are no extra SN parameters
    ↪to loop through
    use_set_peak_magnitude = True,
    peak_abs_mag = -18.6,
    peak_time = 15 #changes ?
)

print(baseline_flux)

```

[4.14881277 4.14881277] mJy

```

[4]: #2. Calculate de-reddened flux density for GRB event
#in this case it is GRB 980425 but still need to calculate
#pass in values defined above from Jillions data

#to get a_v from E(B-V)
a_v = target_milky_way_ebv * 3.1

#convert AB magnitude of arbitrary GRB event --> flux density (mJy)
def calc_flux_density_from_ABmag(AB_mag):
    """
    Calculate flux density from AB magnitude assuming monochromatic AB filter

    :param magnitudes: AB magnitude values
    :return: flux density
    """
    return (AB_mag * uu.ABmag).to(uu.mJy)

flux_density_event = calc_flux_density_from_ABmag(AB_mag)
print(f"This is the flux density of GRB event without extinction correction:
    ↪{flux_density_event:.5f}")

#de-redden flux density, account for extinction if necessary

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dereddened_flux_event = extinction.remove(fitzpatrick99(wavelength, a_v, 3.
    ↪1),flux_density_event)
print(f"This is the extinction corrected GRB event flux density found using
    ↪fitzpatrick99: {dereddened_flux_event[0]:.5f}")

```

This is the flux density of GRB event without extinction correction: 7.04693 mJy
 This is the extinction corrected GRB event flux density found using fitzpatrick99: 8.79830 mJy

```

[5]: # 3. Take the ratio
flux_ratio = dereddened_flux_event / baseline_flux

print(flux_ratio[0])

#should be f_98 = 1 for 1998bw

```

2.120678699823292

```

[6]: #try different data at a later time
AB_mag = 15.96 - 0.09 #B band correction
#have to use two frequencies or two time points - pass in the same time point
    ↪twice
target_obs_times = np.array([37.46000000011176, 37.46000000011176]) #days in
    ↪observer frame
target_redshift = 0.0085
target_frequencies = 6.86300e+14 # B band
wavelength = np.array([4371.07000]) #angstroms
target_milky_way_ebv = 0.059

#interpolate sncosmo model
baseline_flux = func(
    time=target_obs_times,
    redshift=target_redshift,
    model_kwargs=None,
    sncosmo_model='v19-1998bw-corr',
    output_format='flux_density', # Returns mJy
    frequency=target_frequencies,
    mw_extinction=True,           # Add MW dust for YOUR target
    ebv_mw=target_milky_way_ebv, # Specific to your GRB location
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    use_set_peak_magnitude = True,
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    peak_time = 15 #changes ?
)

```

```

#to get a_v from E(B-V)
a_v = target_milky_way_ebv * 3.1

#convert AB magnitude of arbitrary GRB event --> flux density (mJy)
def calc_flux_density_from_ABmag(AB_mag):
    """
    Calculate flux density from AB magnitude assuming monochromatic AB filter

    :param magnitudes: AB magnitude values
    :return: flux density
    """
    return (AB_mag * uu.ABmag).to(uu.mJy)

flux_density_event = calc_flux_density_from_ABmag(AB_mag)

#de-redden flux density, account for extinction if necessary
dereddened_flux_event = extinction.remove(fitzpatrick99(wavelength, a_v, 3.
    ↪1),flux_density_event)

flux_ratio = dereddened_flux_event / baseline_flux

print(flux_ratio[0])

#is code just factor of 2 out for everything ?

```

2.052274769205968

```

[7]: #plot this to see what is actually going on:
freq = target_frequencies
redshift = target_redshift
times = np.linspace(0.01,90,200)
time = target_obs_times

sn_1998bw = func(time=times,
    redshift=redshift,
    model_kwargs=None,
    sncosmo_model='v19-1998bw-corr',
    output_format='flux_density', # Returns mJy
    frequency=target_frequencies,
    mw_extinction=True,           # Add MW dust for YOUR target
    ebv_mw=target_milky_way_ebv, # Specific to your GRB location
    model_kwarg_names=[],
    use_set_peak_magnitude = True,
    peak_abs_mag = -18.6,
    peak_time = 16 #shifts pak time to where it actually is
)

```

```

#now use this as the SN1998bw lightcurve and the above as the GRB event with
↳redshift defined
sn_1998bw1= func(time=times,
    redshift=0.0085,
    model_kwargs=None,
    sncosmo_model='v19-1998bw-corr',
    output_format='flux_density', # Returns mJy
    frequency=target_frequencies,
    mw_extinction=True,           # Add MW dust for YOUR target
    ebv_mw=target_milky_way_ebv, # Specific to your GRB location
    model_kwarg_names=[],
    use_set_peak_magnitude = True,
    peak_abs_mag = -18.6,
    peak_time = 16 #shifts pak time to where it actually is
)

plt.semilogy(times,sn_1998bw,'-b')
plt.semilogy(times,sn_1998bw1, '-r')

plt.show()

#on top of each other which is correct - can i plot Jillions data over this and
↳then match it up to standardise?

```



